Intro

dpkg

Debian Package Manager

- low level package manager
- used to install, configure, remove and query packages
- does not handle dependencies and has no knowledge of repositories

apt

Advanced Package Tool

- high level package manager
- automatically resolves dependencies
- gets packages from repositories

dpkg (low level package manager)

Debian Package Manager
syntax:
`dpkg [options][action] [package]`

- Because dpkg can take a package filename as input, it's the preferred way of installing a package you have already downloaded
- Doesn't upgrade packages, you have to remove the old one and install the one

action	
`-i` `--install`	install
`--configure`	reconfigure an installed package - runs install script to set options
`r` `--remove`	remove package but leave config files
`-P` `--purge`	remove package and config files
`--get-selections`	display installed packages
`-p` `--print-avail`	display info about installed package
`-I` `--info`	display info about uninstalled package
`-l pattern` `--list patter`	list installed packages whose names match *pattern*
`-L` `listfiles`	list files associated with a package
`-S pattern` `--search pattern`	find packages that own the file *pattern*
`-C` `--audit`	find partially installed packages and suggest what to do with them

option	used with action	
`--no-act`	-i, -r	checks for dependency conflicts without installing or removing
`-G`	-i	don't install package if new version of same package is installed
`-E` `--skip-same-version`	-i	don't install package if same version of package already installed

apt

apt is a newer tool
it combines **apt-get**, **apt-cache** and **apt-config**

- apt-get lets you install, remove... packages
- apt-cache shows you info from the package database

apt-get

Repositories list
/etc/apt/sources.list

syntax
`apt-get [options][command][package]`
usually you won't use options

command	
`install`	
`remove`	
`purge`	removes package and configuration files
`update`	says how many packages can be upgrades
`upgrade`	upgrade all packages
`dist-upgrade`	upgrade package if it doesn't break a dependency
`download`	download package only
`clean`	clear local cache (not database) of downloaded packages

option	used with	
`-d` `--download-only`	upgrade, dselect-upgrade, install, source	downloads packages without installing
`-f` `--fix-broken`	install, remove	fix system on which dependencies are unsatisfied

apt-cache

command	
`showpkg`	show package info
`stats`	package stats - how many you've installed etc
`unmet`	if a program has missing libraries/files
`depends [package]`	show package dependencies
`pckgames`	shows all installed packages if followed by a string, only those that begin with the string

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Reconfiguring packages

Install scrips often ask a handful of questions
to re-customise run `dpkg-reconfigure [package]`

Debian source packages

Files that are used to build the binary package that you interact with contains:

- tarball
- patch file - used to modify source code
- .dsc file - contains a digital signature to verify the files

You can compile these to create a Debian binary package

tar

`.tar`
stand for **T**ape **AR**chiver
bundles files and directories together but doesn't compress them
`tar something.tar file1 file2 dir1`

tarball

A compressed tar file (like a zip file)
`.tar.gz` (**tar** + **gzip** compression)